

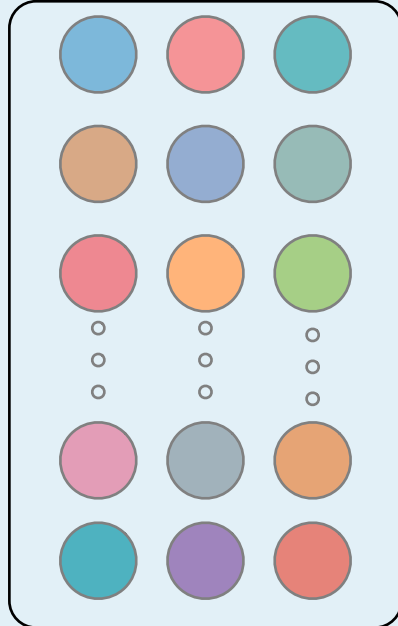
Module 1: Reference Learning from scRNA-seq

scRNA-seq
count matrix
(cells x genes)

Cell-type
labels

Slot Decoder Network

Learnable
prototype slots $\mathbf{S}$



Shared
Decoder
MLP

Negative Binomial
Likelihood loss

Reconstructed
Mean

Learned
signature
matrix $\mathbf{B}$

Module 2: Spatial Deconvolution of ST Data

Signature
matrix $\mathbf{B}$
(from module 1)

ST expression matrix
(spots x genes)

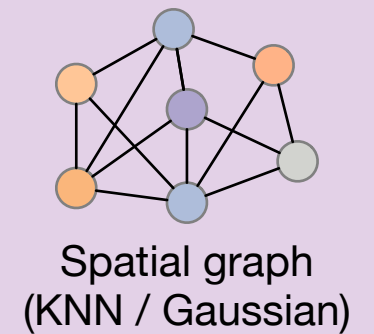
library-size
normalization

Weighted NNLS

Step A
(Initialization,
Non-spatial)

Initial proportions $\mathbf{V}_{init}$

Spatial
coordinates
(x, y)



Spatial graph
(KNN / Gaussian)

Step B (Spatial
refinement)

KL divergence
reconstruction loss

Optimization loop

Spatial smoothness
regularization

Refined cell-type proportions $\mathbf{V}_{final}$